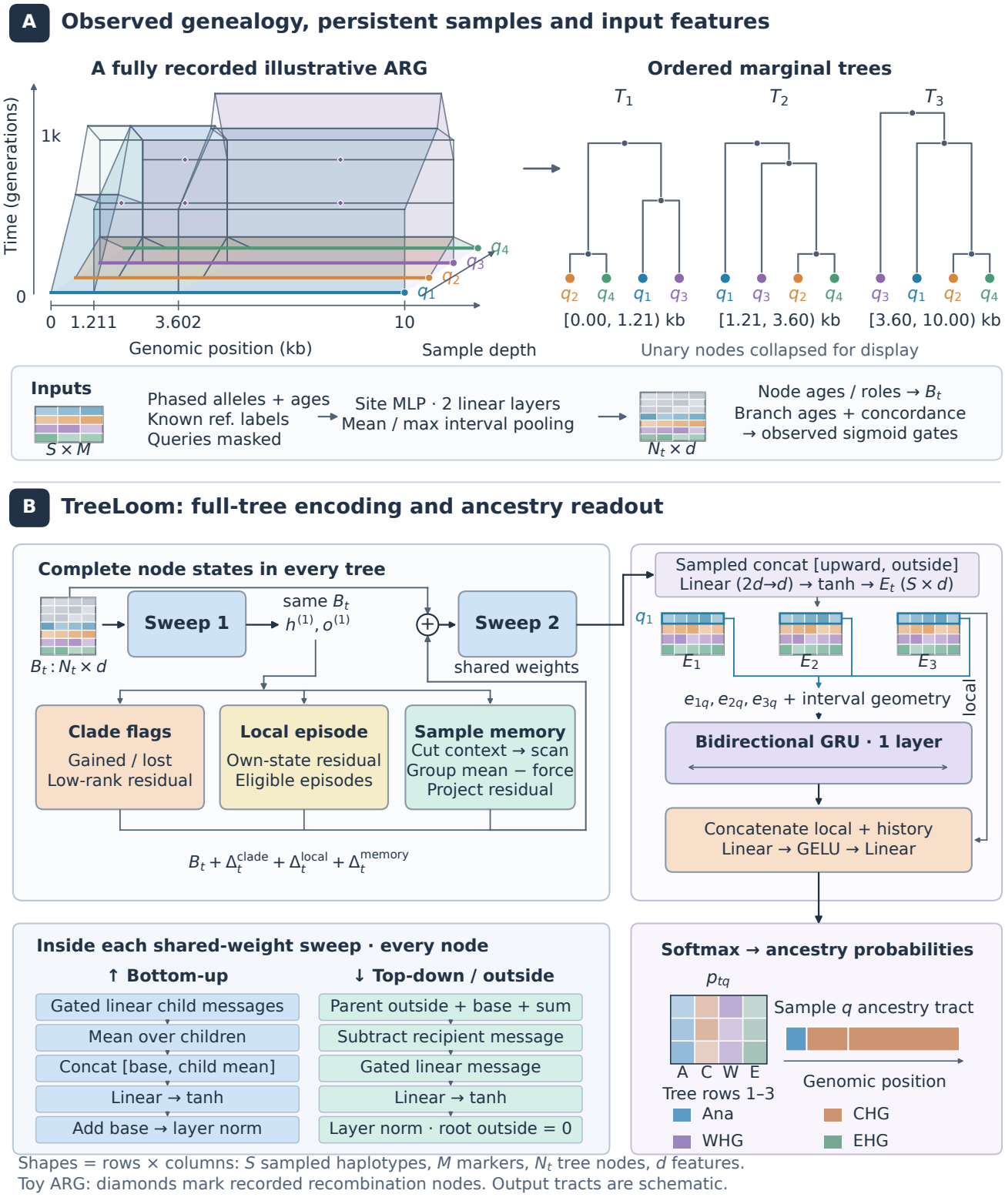


Figure 1: TreeLoom architecture. **A:** An illustrative ARG with recorded recombination events yields three marginal trees. Sheets follow genomic edges. Tip colors identify samples. **B:** Two shared whole-tree sweeps, local residuals, sample memory and a one-layer bidirectional GRU feed the ancestry classifier. Outputs are schematic.